

Phys 115B HW 1

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Question 1. *You just know the answer*

A common problem in QM is “find the energies of the following Hamiltonian.” The idea is to do as little math as possible but to build intuition. Describe the lowest several eigenenergies of the following Hamiltonians. It is up to you to decide what value of “several” best illustrates what is going on. Answers are short.

Example: What are the energies of the two separate simple harmonic oscillators, each with frequency ω ?

Answer: The oscillators have energies

$$E_1 = \hbar\omega(1/2 + n_1), \quad n_1 = [0, 1, 2, 3, \dots]$$

$$E_2 = \hbar\omega(1/2 + n_2), \quad n_2 = [0, 1, 2, 3, \dots]$$

so the combined oscillators have energy

$$E = E_1 + E_2 = \hbar\omega(1 + n_1 + n_2) = \hbar\omega[1, 2, 2, 3, 3, 3, 4, 4, 4, 4, \dots]$$

1. Particle of mass m in one dimensional box of width a .
2. Particle of mass m in two dimensional box of width a and length b .
3. Two disconnected particles, each of mass m , one in a 1D box of width a and the other in a 1D box of width b .
4. Particle of mass m confined to a circular ring of circumference a .
5. Particle of mass m in a one dimensional potential $V = \frac{1}{2}m\omega^2x^2$.
6. Particle of mass m in a two dimensional potential $V = \frac{1}{2}m\omega^2(x^2 + y^2)$.

Proof.

□

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Question 2. Two-body to one-body (Griffiths 5.1 in both editions)

Typically, the interaction potential depends only on the vector $\mathbf{r} := \mathbf{r}_1 - \mathbf{r}_2$ separating the two particles. In that case the Schrödinger equation separates, if we change variables from $\mathbf{r}_1, \mathbf{r}_2$ to $\mathbf{r}, \mathbf{R} := (m_1\mathbf{r}_1 + m_2\mathbf{r}_2)/(m_1 + m_2)$ (the center of mass).

(a) Show that $\mathbf{r}_1 = \mathbf{R} + (\mu/m_1)\mathbf{r}$, $\mathbf{r}_2 = \mathbf{R} - (\mu/m_2)\mathbf{r}$, and $\nabla_1 = (\mu/m_2)\nabla_R + \nabla_r$, $\nabla_2 = (\mu/m_1)\nabla_R - \nabla_r$, where

$$\mu := \frac{m_1 m_2}{m_1 + m_2}$$

Proof.

□

3 (ND)

Question 3. SHO in 3D (Modified Griffiths 3rd ed. 4.46, or 2nd ed. 4.39)

Consider a particle of mass m for which the potential is

$$V(r) = \frac{1}{2}m\omega^2 r^2$$

- (a) Using separation of variables in Cartesian coordinates, show that this factorizes into a sum of three 1D harmonic oscillators, and use your knowledge of 1D SHO to determine the allowed energies.
- (b) Determine the degeneracy $d(n)$ (i.e. the number of states with the same energy as the n th energy eigenvalue E_n).
- (c) As you hopefully found in part (b), the ground state $\psi_0(x, y, z)$ is non-degenerate. However the first excited energy eigenvalue E_1 is threefold degenerate, with eigenstates:

$$\psi_{1,0,0}(x, y, z) = \tilde{\psi}_1(x)\tilde{\psi}_0(y)\tilde{\psi}_0(z)$$

$$\psi_{0,1,0}(x, y, z) = \tilde{\psi}_0(x)\tilde{\psi}_1(y)\tilde{\psi}_0(z)$$

$$\psi_{0,0,1}(x, y, z) = \tilde{\psi}_0(x)\tilde{\psi}_0(y)\tilde{\psi}_1(z)$$

where $\tilde{\psi}_i$ denotes an eigenstate of the 1D harmonic oscillator. Are any of the linear combinations

$$\frac{1}{\sqrt{3}}(\psi_{1,0,0} + \psi_{0,1,0} + \psi_{0,0,1}), \quad \frac{1}{\sqrt{2}}(\psi_{1,0,0} + \psi_{0,1,0}), \quad \frac{1}{\sqrt{2}}(\psi_{1,0,0} + i\psi_{0,0,1})$$

also energy eigenstates? In each case, why or why not?

Proof.

□

Question 4. 1D to 3D (Griffiths 4.1, same in both editions)

- (a) Work out all of the **canonical commutation relations** for components of the operators $\mathbf{r}$ and $\mathbf{p}$: $[\hat{x}, \hat{y}]$, $[\hat{x}, \hat{p}_y]$, $[\hat{x}, \hat{p}_x]$, $[\hat{p}_y, \hat{p}_z]$, and so on.
- (b) Confirm the three-dimensional version of **Ehrenfest's Theorem**,

$$\frac{d}{dt} \langle \mathbf{r} \rangle = \frac{1}{m} \langle \mathbf{p} \rangle, \quad \frac{d}{dt} \langle \mathbf{p} \rangle = \langle -\nabla V \rangle$$

- (c) Formulate **Heisenberg's Uncertainty Principle** in three dimensions.

Proof.

- (a) Given $j, k \in \{x, y, z\}$, the action of $[\hat{j}, \hat{k}]$ on the states in position basis are as follow:

$$[\hat{j}, \hat{k}] |\psi\rangle = (\hat{j}\hat{k} - \hat{k}\hat{j}) \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 = (j\hat{k} \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 - \hat{k} \int_{\mathbb{R}^3} j\psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3) \quad (1)$$

$$= \int_{\mathbb{R}^3} jk\psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 - \int_{\mathbb{R}^3} kj\psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 = 0 \quad (2)$$

Similarly, the action of $[\hat{p}_j, \hat{p}_k]$ on the states as follow:

$$[\hat{p}_j, \hat{p}_k] |\psi\rangle = -\hbar^2 \left(\frac{\partial}{\partial j} \frac{\partial}{\partial k} - \frac{\partial}{\partial k} \frac{\partial}{\partial j} \right) \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 \quad (3)$$

$$= -\hbar^2 \left(\frac{\partial}{\partial j} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial k} |\bar{r}\rangle d\bar{r}^3 - \frac{\partial}{\partial k} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial j} |\bar{r}\rangle d\bar{r}^3 \right) \quad (4)$$

$$= -\hbar^2 \left(\int_{\mathbb{R}^3} \frac{\partial}{\partial j} \left(\frac{\partial \psi}{\partial k} \right) |\bar{r}\rangle d\bar{r}^3 - \int_{\mathbb{R}^3} \frac{\partial}{\partial k} \left(\frac{\partial \psi}{\partial j} \right) |\bar{r}\rangle d\bar{r}^3 \right) = 0 \quad (5)$$

Which, it's due to the fact that the mixed partial derivatives are equal if the function is smooth / at least twice continuously differentiable.

Now, the action of $[\hat{j}, \hat{p}_k]$ is as follow:

$$[\hat{j}, \hat{p}_k] |\psi\rangle = -i\hbar \left(\hat{j} \frac{\partial}{\partial k} - \frac{\partial}{\partial k} \hat{j} \right) \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 \quad (6)$$

$$= -i\hbar \left(\hat{j} \int_{\mathbb{R}^3} \frac{\partial \psi}{\partial k} |\bar{r}\rangle d\bar{r}^3 - \frac{\partial}{\partial k} \int_{\mathbb{R}^3} j\psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 \right) \quad (7)$$

$$= -i\hbar \left(\int_{\mathbb{R}^3} j \frac{\partial \psi}{\partial k} |\bar{r}\rangle d\bar{r}^3 - \int_{\mathbb{R}^3} \left(\frac{\partial j}{\partial k} \psi(\bar{r}) + j \frac{\partial \psi}{\partial k} \right) |\bar{r}\rangle d\bar{r}^3 \right) \quad (8)$$

$$= -i\hbar \left(\int_{\mathbb{R}^3} j \frac{\partial \psi}{\partial k} |\bar{r}\rangle d\bar{r}^3 - \delta_{jk} \int_{\mathbb{R}^3} \psi(\bar{r}) |\bar{r}\rangle d\bar{r}^3 - \int_{\mathbb{R}^3} j \frac{\partial \psi}{\partial k} |\bar{r}\rangle d\bar{r}^3 \right) \quad (9)$$

$$= i\hbar \delta_{jk} |\psi\rangle \quad (10)$$

So, as a result, we get:

$$[\hat{x}, \hat{y}] = 0, \quad [\hat{x}, \hat{p}_y] = 0, \quad [\hat{x}, \hat{p}_x] = i\hbar, \quad [\hat{p}_y, \hat{p}_z] = 0 \quad (11)$$

- (b) If limit to one dimension (pick $j \in \{x, y, z\}$), then Ehrenfest's theorem in one dimension guarantees that $\frac{d}{dt} \langle j \rangle = \frac{1}{m} \langle p_j \rangle$ and $\frac{d}{dt} \langle p_j \rangle = \left\langle -\frac{\partial V}{\partial j} \right\rangle$. Hence, the three-dimensional version follows, since now if $\mathbf{p}$ and $\mathbf{r}$ are $(\hat{p}_x, \hat{p}_y, \hat{p}_z)$ and $(\hat{x}, \hat{y}, \hat{z})$ respectively, one has:

$$\frac{d}{dt} \langle \mathbf{r} \rangle = \frac{d}{dt} (\langle x \rangle, \langle y \rangle, \langle z \rangle) = \frac{1}{m} (\langle p_x \rangle, \langle p_y \rangle, \langle p_z \rangle) = \frac{1}{m} \langle \mathbf{p} \rangle \quad (12)$$

$$\frac{d}{dt} \langle \mathbf{p} \rangle = \frac{d}{dt} (\langle p_x \rangle, \langle p_y \rangle, \langle p_z \rangle) = \left(\left\langle -\frac{\partial V}{\partial x} \right\rangle, \left\langle -\frac{\partial V}{\partial y} \right\rangle, \left\langle -\frac{\partial V}{\partial z} \right\rangle \right) = \langle -\nabla V \rangle \quad (13)$$

Since $-\nabla V = \left(-\frac{\partial V}{\partial x}, -\frac{\partial V}{\partial y}, -\frac{\partial V}{\partial z} \right)$.

- (c) Given $j, k \in \{x, y, z\}$, since $[\hat{j}, \hat{p}_k] = i\hbar\delta_{jk}$ (verified in part (a)), then by the generalized uncertainty principle, one has:

$$\sigma_j \sigma_{p_k} \geq \frac{1}{2} \left| [\hat{j}, \hat{p}_k] \right| = \frac{\hbar}{2} \delta_{jk} \quad (14)$$

Hence, if $j = k$, then $\sigma_j \sigma(p_j) \geq \frac{\hbar}{2}$, yet if $j \neq k$, one has $\sigma_j \sigma_{p_k} \geq 0$, which imposes no restriction.

□

5 (ND)

Question 5. Cubical well (Griffiths 4.2, same in both editions)

Use separation of variables in cartesian coordinates to solve the infinite cubical well (or “particle in a box”):

$$V(x, y, z) = \begin{cases} 0 & x, y, z \in [0, a] \\ \infty & \text{otherwise} \end{cases}$$

- (a) Find the stationary states, and the corresponding energies.
- (b) Call the distinct energies $E_1, E_2, E_3, \dots$ in order of increasing energy. Find E_1, E_2, E_3, E_4, E_5 , and E_6 . Determine their degeneracies (that is, the number of different states that share the same energy).
- (c) What is the degeneracy of E_{14} , and why is this case interesting?

Proof. Suppose

□

6 (check if the normalization claim is correct)

Question 6. Practice with Spherical Harmonics (Griffiths 4.4 in 3rd edition, 4.3 in 2nd edition)

Show that

$$\Theta(\theta) = A \ln[\tan(\theta/2)]$$

satisfies the Θ equation for $\ell = m = 0$. This is the unacceptable "second solution" - what's wrong with it?

Note: the equation is of the form

$$\sin(\theta) \frac{d}{d\theta} \left(\sin(\theta) \frac{d\Theta}{d\theta} \right) + [\ell(\ell+1) \sin^2(\theta) - m^2] \Theta = 0$$

Proof. Given $\ell = m = 0$, the given equation reduces to:

$$\sin(\theta) \frac{d}{d\theta} \left(\sin(\theta) \frac{d\Theta}{d\theta} \right) = 0 \quad (15)$$

Hence, one gets $\sin(\theta) \frac{d\Theta}{d\theta} = A_1$ for some constant $A_1 \in \mathbb{C}$.

Now, since this implies $\frac{d\Theta}{d\theta} = \frac{A_1}{\sin(\theta)} = A_1 \csc(\theta)$, then one gets the following:

$$\Theta = \int A_1 \csc(\theta) d\theta = -A_1 \ln \left| \tan \left(\frac{\theta}{2} \right) \right| + A_2 \quad (16)$$

Which, for $\theta \in (0, \pi)$, $\tan(\theta/2) > 0$, hence $\Theta = A \ln[\tan(\theta/2)] + B$ for some $A, B \in \mathbb{C}$.

However, this is not an acceptable solution, since the integration of $|\Theta|^2$ over $\theta \in (0, \pi)$ in fact diverges, showing that Θ is not normalizable. $\square$